Pre-Analysis Plan (PAP)

Project Title: Causal Effects of Having Daughters on Gender Attitudes and Behaviors

Data Source: SOEP (German Socio-Economic Panel)

1. Introduction and Motivation

This pre-analysis plan outlines the methodology to test four causal hypotheses regarding the effect of having daughters on gender-related attitudes, political preferences, and intra-household behaviors. The aim is to pre-specify the empirical strategy, variable definitions, and regression models for transparency and credibility.

2. Data and Sample

Data Source: SOEP (longitudinal German household panel).
Period: Tentatively 2000–2022 (depending on variable availability).

Inclusion Criteria:

- Being more than 15 years old
- In couple & having kids for H3 (with valid parid)

Exclusion Criteria:

- Only participants of private households are included
- Respondents without children in H3
- Missing values on key variables

3. Hypotheses and Analytical Plans

H1. Having daughters increases support for gender equality (attitudes).

Dependent Variable: gender_eq_mean1

Items used (each on a Likert scale from 1 to 5, where 1 = strongly disagree, 5 = strongly agree):

1. plh0395i03: *"It is completely natural that there are people with both male and female traits."*
2. plh0395i04: *"Being assigned to a gender is not essential for a person's development."*

3. plh0395i05: *"Schools and kindergartens should encourage children to engage in activities that are not typical for their gender."*
4. plh0395i06: *"Same-sex couples wishing to have children should be accepted in our society."*

Independent Variable: has_daughter

- Dummy = 1 if respondent has at least one daughter among children kidsex01–kidsex19

Control Variables:

- nb_kid: number of children
- age, age2: age and age squared
- plh0182 : current life satisfaction
- hghinc: household income
- pgfamstd: marital status
- pgpsbil: level of education
- pgemplst: employment status
- syear: survey year fixed effects
- sex: respondent gender
- sampreg: region (East/West)
- migback: migration background

Regression Equation:

We will use an OLS regression model to examine how having at least one daughter (has_daughter) relates to the average gender equality attitude score (gender_eq_mean1, gender_eq_mean2), controlling for the total number of children (nb_kid), age and its squared term (age, age2), current life satisfaction (plh0182), household income (hghinc), family standard classification (pgfamstd), educational attainment (pgpsbil), employment status (pgemplst), gender (sex), sampling region (sampreg), and migration background (migback). The analysis includes only respondents aged over 15 and uses robust standard errors. The model aims to isolate the association between having daughters and progressive gender equality attitudes, while adjusting for demographic and socioeconomic factors. As there is a single wave of respondents, we use unclustered robust standard errors.

The following equation is tested :

$$\text{gender_eq_mean2_it} = \beta_0 + \beta_1 \cdot \text{has_daughter_it} + \gamma' \cdot \text{X_it} + \varepsilon\text{_it}$$

with X_it the set of controls and ε_it the error term. The coefficient of interest is β_1. To validate the hypothesis, β_1 should be positive and significant at the 0.05 level ($p<0.05$).

H2. Having daughters increases left-leaning political preferences.

- **Dependent variable**: left = political ideology dummy (recoded so that higher = more left) based on plh0012_h
- Left =1 if political preferences are for the following parties [1] SPD, [5] Bündnis 90 / Die Grünen, [6] Die Linke, [9] SPD/Buendnis90/Gruene, [16] Buendnis90/Gruene/Die Linke, [17] SPD/Die Linke, [19] Die Linke /DVU(Rep/NPD), [24] FDP/Die Linke, [26] Piratenpartei
- **Same controls as H1**

The OLS regression model investigates whether having a daughter (has_daughter) is associated with their political ideology (left). It controls for the total number of children (nb_kid), age (age), and age squared (age2) to account for non-linear age effects. Socio-economic controls include current life satisfaction (plh0182) and household income (hghinc). The model also includes dummies for family status (pgfamstd), education level (pgpsbil), employment status (pgemplst), survey year (syear), sex (sex), federal state (sampreg), and migration background (migback). The analysis is limited to individuals older than 15 (if age > 15), and standard errors are clustered by individual ID (pid) to account for repeated observations over time. Robust standard errors are used to correct for heteroskedasticity.

The following equation is tested:

$$left_it = \beta_0 + \beta_1 \cdot has_daughter_it + \gamma' \cdot X_it + \varepsilon_it$$

with X_it the set of controls and ε_it the error term. The coefficient of interest is β_1. To validate the hypothesis, β_1 should be positive and significant at the 0.05 level ($p<0.05$).

H3. Having daughters increases gender equality in household behavior

- **Outcomes**:
  - diff_childcare = husband's avg time on childcare − wife's avg time
- **Construction**:
  - Keep individuals with valid parid (partner ID)
  - Construct couple identifier: couple_id = min(pid, parid)
  - Reshape to wide by sex to align male and female partner on same row
  - Compute within-couple difference between female and male in time spent on childcare, and then compute the difference between the score for a given couple *j* and the annual average. Higher score indicates relatively greater involvement of man in childcare compared to other couples in a given year.
- Same controls as H1 and H2

We estimate an OLS regression models to assess whether having daughters is associated with greater gender equality in household childcare behaviors. The dependent variable measurse the within-couple difference in time spent on childcare relative to the annual

average. Positive values indicate greater involvement of men in childcare compared to other couples. These variables were constructed by subtracting the yearly mean difference in childcare time (`mean_diff_childcare`) from each couple's raw difference (`diff_childcare`), allowing us to analyze relative deviations from the yearly norm. The key independent variable is has_daughter, a binary indicator equal to 1 if the household has at least one daughter. Controls include the total number of children (nb_kid), respondent's age and its square (age, age2), household income (hghinc), and life satisfaction (plh0182). Additionally, individual-level covariates such as family status (pgfamstd), employment status (pgemplst), and education level (pgpsbil) are included. Fixed effects for survey year (syear), gender (sex), region (sampreg), and migration background (migback) are also controlled for. Standard errors are clustered at the individual level (pid) and are robust to heteroskedasticity.

The following equation is tested:

$$\text{diff_childcare}_{it} = \beta_0 + \beta_1 \cdot \text{has_daughter}_{it} + \gamma' \cdot X_{it} + \varepsilon_{it}$$

with X_{ijt} the set of controls and ε_{ijt} the error term. The coefficient of interest is β_1. To validate the hypothesis, β_1 should be positive and significant at the 0.05 level ($p<0.05$).

H4. The effect of daughters is stronger for men

- The model is similar to the one tested in H2 bud adds an interaction between `sex` (coded 1 = male, 2 = female) and `has_daughter`
- We look at the coefficient β_3 of the interaction between `sex` =1 and `has_daughter` to test the hypothesis.
- Same controls as H1, H2 and H3.

The model examines the effect of having at least one daughter (has_daughter) on political ideology (left), testing whether this effect differs by gender through an interaction term between sex and has_daughter. It controls for the total number of children (nb_kid), age (age) and age squared (age2), as well as socio-economic variables (plh0182, hghinc), and includes fixed effects for family standard (pgfamstd), bilateral relations (pgpsbil), employment status (pgemplst), survey year (syear), sample region (sampreg), migration background (migback), and gender (sex). The significance and sign of the interaction term 1.sex#i.has_daughter indicate whether having daughters impacts men and women differently, thus testing hypothesis H4 that this effect is stronger for men. The analysis is limited to individuals older than 15 (if age > 15).

The following equation is tested:

$$\text{left}_{it} = \beta_0 + \beta_1 \cdot \text{has_daughter}_{it} + \beta_2 \cdot \text{sex(=1)}_{it} + \beta_3 \cdot (\text{has_daughter}_{it} \times \text{sex (=1)}_{it} + \gamma' \cdot X_{it} + \varepsilon_{it}$$

with X_{it} the set of controls and ε_{it} the error term. The coefficient of interest is β_3. To validate the hypothesis, β_1 should be positive and significant at the 0.05 level ($p<0.05$).

4. Methodology

- All models estimated via OLS regressions.
- We use robust standard errors, clustered on `pid` if repeated observations (in H2, H3 & H4)
- Missing or implausible values (`< 0`) are set to missing before aggregation
- A hypothesis is considered supported if β_1 (or β_3 in H4) is statistically significant and in the predicted direction.
- All models include extensive socio-demographic controls.